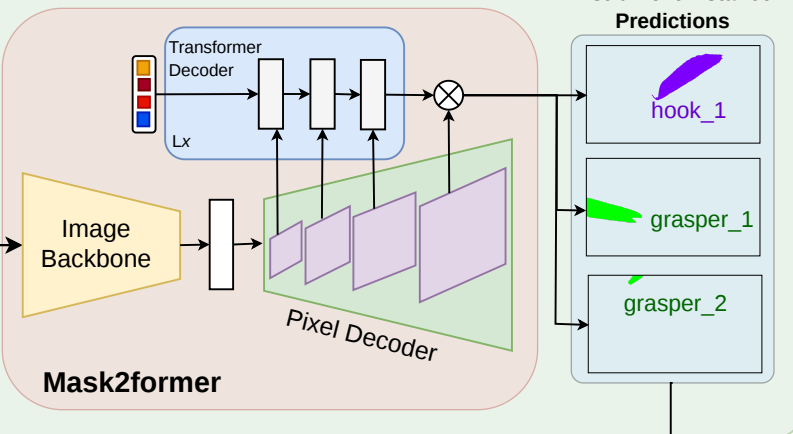
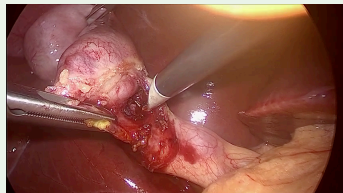


Stage 1: Instrument Instance Segmentation

Input Image

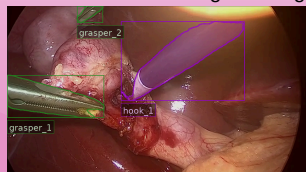


Mask-based Visual Prompt Construction

RGB Image + Translucent Masks + Bounding Box + Instance Identifiers

Inputs to Qwen3-VL

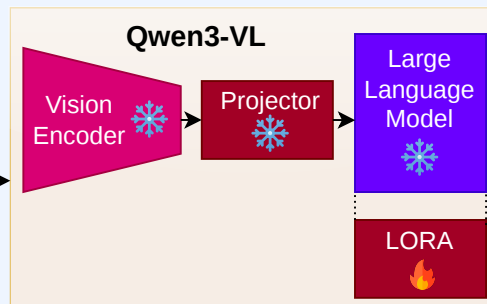
* Instance Overlaid Surgical Image



Text Prompt

For each instrument instance, assign exactly one (instrument, verb, target) triplet using visual evidence.
...

Stage 2: Multimodal Interaction Prediction



Predicted Grounded Surgical Triplets

Mask	Instrument Instance	Verb	Target
	hook_1	dissect	cystic duct
	grasper_1	retract	cystic duct
	grasper_2	retract	gallbladder

interactions